

Group Project (28/07/2026)

Start Assignment

- Due 28 Jul by 23:59
- Points 20
- Submitting a text entry box or a file upload
- Available 14 Jul at 21:30 - 28 Jul at 23:59



Group Project: Applied Natural Language Processing



Context

The goal of this project is to apply the concepts and techniques covered in this course to design and implement a small but complete Natural Language Processing (NLP) application.

Students will work in groups to build an end-to-end NLP pipeline that addresses a real-world or business-relevant problem. The focus is on demonstrating how NLP techniques can be used to extract, process, and generate value from textual data in practical contexts.

The emphasis is on correct application of NLP methods, coherent pipeline design, and meaningful interpretation of results, rather than on system complexity or production-ready solutions.



Objectives

By completing this project, students will be able to:

- Apply the full NLP workflow, from data acquisition through text processing and representation to task-level modelling;
- Identify and formulate a real-world or business-relevant problem that can be addressed using Natural Language Processing techniques;

- Implement fundamental NLP techniques, including named entity recognition, keyword extraction, text classification, sentiment analysis, and summarization;
 - Optionally integrate Large Language Models (LLMs) via APIs or platforms such as OpenAI to enhance or extend the solution;
 - Interpret and analyse the outputs of NLP models in a meaningful and contextualized way;
 - Communicate clearly the design, implementation, and results of an NLP-based solution.
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Team Formation

- Groups of up to **4-5 students**
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Project Requirements

Each group must develop a **simple functional NLP application** addressing a real-world or business-relevant problem.

1. Problem Definition

- Clearly describe the problem and its context (real-world or organizational/business setting);
- Explain why NLP is appropriate for addressing the problem.

2. Data

- Use of real data (at least one source):
 - web (APIs or scraping),
 - documents (PDF/text),
 - or provided datasets.

3. NLP Pipeline

- The solution must include a basic NLP pipeline covering the studied workflow.

4. Optional LLM Component

- Integration of a Large Language Model (LLM) such as text generation, summarization, etc.

5. Prototype Implementation

- A working prototype demonstrating the solution, implemented using Streamlit, Hugging Face Spaces, or similar tools.
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Deliverables

Each group must submit:

1. Presentation (PDF)

- A short slide deck (**max. 10 slides**) including:
 - Problem and context (including business/real-world relevance)
 - Data sources
 - NLP pipeline overview
 - Methods used
 - Optional LLM integration
 - Key results and interpretation
 - Limitations and possible improvements

2. Prototype

- A functional demo accessible via:
 - Streamlit app, Hugging Face Space, or equivalent
 - A link must be provided in the presentation (pdf)

3. Short Video Presentation

- Maximum **3 minutes**
- Clear explanation of:
 - problem
 - solution
 - demonstration of the prototype
- Focus on clarity, not technical depth

- A link must be provided in the presentation (pdf)
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Key Notes

- Simplicity is preferred over complexity
 - A well-executed simple pipeline is better than a complex but incomplete system
 - The goal is to demonstrate understanding of NLP concepts applied to real data
 - Use of LLMs is optional and should complement, not replace, the NLP pipeline
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Evaluation Criteria

Evaluation will be simple and qualitative, focusing on creativity and clarity of thinking. The main goal is for students to apply what they have learned in a meaningful way, rather than to deliver a polished product. Each group will be evaluated based on:

1. Problem Relevance & Interpretation (40%)

- **Clarity of the problem definition.**
Is the problem clearly stated and well motivated?
- **Quality of interpretation of results.**
Are the outputs of the NLP approach meaningfully analysed and explained?
- **Relevance of the NLP approach.**
Is NLP an appropriate and effective choice for the proposed problem?

2. Prototype Implementation (40%)

- **Functionality of the solution.**
Does the prototype work as intended and demonstrate the core idea?
- **Correct and coherent use of NLP techniques.**
Are the selected methods appropriate and correctly applied?

- **Use of tools and libraries.**

Are tools and frameworks (e.g. Python libraries or APIs) used appropriately?

- **Clarity and usability of the demo.**

Is the prototype easy to understand and interact with?

3. **Communication (20%)**

- **Clarity of the slides.**

Is the presentation concise, well structured, and easy to follow?

- **Quality of the video explanation.**

Is the solution clearly explained and well communicated?